Image segmentation and classification of skin lesions

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***Abstract*—With the massive volume of video and image data flowing worldwide, Computer Vision has emerged as a critical skill to possess. In this work several techniques, like Otsu's thresholding, are employed to segment Melanoma and Melanocytic Nevus images from the ISIC 2019 challenge dataset. Melanomas can be recognised by a few important details which are encoded in different features of the images. These are extracted and fed to a Random Forest algorithm to classify the images in the two types of lesion. The final classification has an accuracy of $\sim 75\%$ for both classes, proving the pipeline to be adequate for this type of images.**

***Index Terms*—Segmentation, Random Forest, Melanoma, Melanocytic Nevus**

I. Introduction

The goal of the project is to apply computer vision algorithms to isolate lesions from the ISIC 2019 challenge dataset[1], then classify them using machine learning. The dataset comprises 25 331 dermoscopic images divided in nine diagnostic categories among which only two are used:

- Melanoma – 4522 images.
- Melanocytic Nevus – 12 875 images.

To detect a melanoma it is necessary to extract meaningful features to be fed to the machine learning algorithm. These should represent the ABCDE rule of Melanomas:

- Asymmetry
- Border Irregularity
- Color Variegation
- Diameter
- Evolution

For obvious reasons it is not possible to track evolution in this kind of dataset, neither to evaluate the true diameter since there is no guarantuee the images share the same scale (in fact there is evidence they do not). Asymmetry is extracted from the final mask used on the image. Border irregularity is evaluated by applying SIFT which can also help with color variegation. The latter is better represented by the means and standard deviations of the CIELAB channels.

A Random Forest algorithm is deemed suitable to perform the final classification, due to its lightweight and performance.

Section II describes the processing pipeline, whose parameters are described in Section III, along with a brief discussion on discarded methods. The results are shown in Section IV and the report is concluded in Section V.

[1]https://www.kaggle.com/datasets/andrewmvd/isic-2019

II. Methodology

Most of the operations are performed in the CIELAB color space so as to capture perceptually different colors. The steps are illustrated in Fig. 1 on page 3.

A. Preprocessing

In order to remove hair and artifacts, a 'closing' morphological operation is first applied. This consists in appyling first erosion, then dilation on the image. The kernel and number of iterations are empirically selected so as not to lose too many details of the lesion, while being able to remove thin hair and artifacts, and reduce the intensity of thicker hair (1b).

After artifacts removal, CLAHE is applied on the L channel to enhance contrast (1c).

The actual thresholding is the intersection of three masks. The first one is obtained by limiting the H channel of the HSV color space, to keep only brownish to blueish hues ($H < 23$ and $H > 110$), which also helps in removing colored band-aids appearing in some images. A second mask is created from the grayscale image to remove the black silhouette from the dermoscopy instrument present in a large share of the images ($G > 10$). Finally Otsu's thresholding is applied on the L channel.

Otsu's algorithm selects the darkest regions, hence the lesion is not the only considered part of the image. To overcome this issue the contours of the mask are computed, and all the patches whose area are less than a threshold or whose centroid is too close to the border of the image are discarded (1d).

Finally a bilateral filter is applied to reduce noise (1e).

B. Features extraction

1) Asymmetry: asymmetry is measured by five features from the final mask: width and height, and symmetries with respect to the vertical and horizontal axis, and the central point. The symmetries across the axes are obtained by simply flipping the mask about the appropriate axis, and a bitwise AND operation, then the fraction of 'active' pixels is evaluated. The same procedure is applied for the central symmetry, but the flipping is performed across both axes.

2) Border irregularity: in order to assess this feature SIFT algorithm is applied to the masked image (1f). Given the average amount of detected features, keeping all the features' description would possibly overwhelm the machine learning algorithm, hence only the number of the features is passed on.

3) Color variegation: this is primarily evaluated by passing the mean and standard deviation of the CIELAB channels, and partially also by the number of features, as SIFT detects blobs and edges inside the image too.

C. Classification

After the feature extraction, the final classification is done via a random forest algorithm implemented with scikit-learn.

The dataset is divided in a training, validation, and test set in order to select the most appropriate hyperparameters, namely the number of trees, and their maximum depth.

III. Discussion

The main issue with the algorithm is the amount of arbitrarily fixed parameters and operations to overcome specific problems within the images.

A. Parameters

The closing operation is performed with a 3×3 kernel, and two iterations (two rounds of erosion, followed by two of dilation). This takes care of hair with thickness of about $3\,\mathrm{px}$, and attenuates larger hair's intensity. Using more aggressive parameters, as larger kernel or number of iterations would destroy thicker hair, but also too much information of the lesion.

The main CLAHE parameters are `clipLimit=1` and `tileGridSize=(10,10)`, which show to enhance the images by improving contrast without making the lighter regions too dark, as this would affect the subsequent Otsu's thresholding.

The bilateral filter is applied with a neighbourhood diameter of 30, $\sigma_{Color} = 20$ and $\sigma_{Space} = 10$, smoothing regions of the lesion with similar color, without destroying edges between patches of different colors.

B. Random Forest

80% of the data ($13\,917$ images) is used to train the random forest algorithm, approximately 13% (2296 images) is used as a validation set to fix the hyperparameters, and the remaining (1184 images) is used to test the chosen model.

The parameters (50 trees and maximum depth of 7) are chosen so as to obtain satisfactory classification for both classes in the validation set. The weights of the two classes are balanced with respect to their frequencies, since the amount of Melanocytic Nevi is three times the amount of Melanomas, which is why a simple accuracy defined as the fraction of correctly classified samples would not be as significant.

C. Discarded approaches

The first approaches consisted in a naive and crude implementation of thresholding in HSV and CIELAB colorspaces, but they have been soon discarded because of the varying ranges of the images in the dataset. An attempt to equalize the images based on the histogram of a 'typical' image was carried out, but it would alter too much the contents of the images.

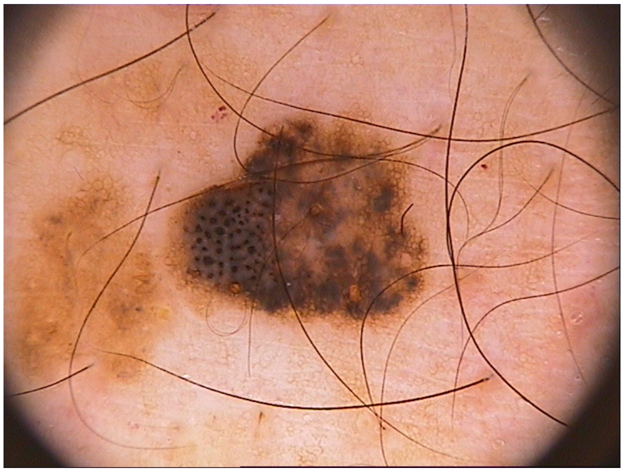
Another way would be implementing K-Means clustering on the intensity of the images, but the final implementation with Otsu's thresholding proved to be faster and more reliable.

An additional attempt has been carried out with watershed—partly implemented as suggested by the OpenCV documentation—but it led to poor results, and the final algorithm was already producing adequate segmentation.

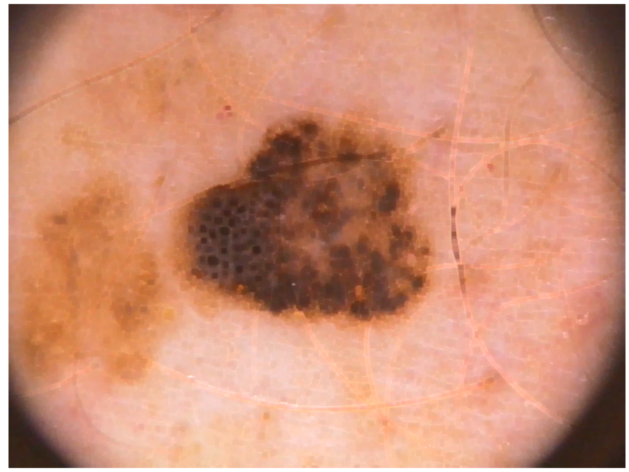
IV. Results

A. Classification

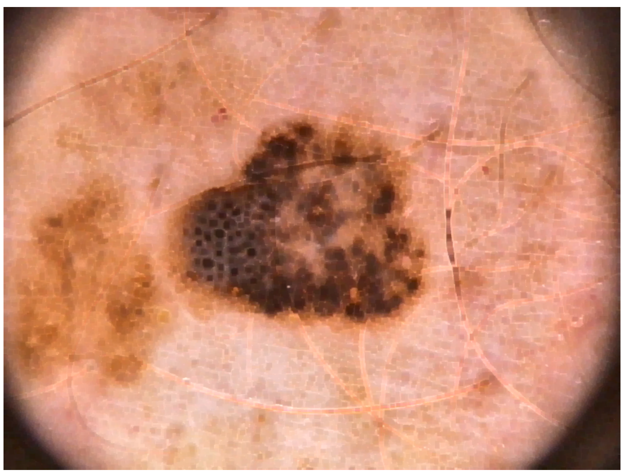
In Tab. 1 the confusion matrix of the best model over the test set is shown. Both classes are correctly identified with $\sim 75\%$ probability. This is coherent with the confusion matrix of the validation set with true positive (NV) probability of $\sim 75\%$, and true negative (MEL) $\sim 74\%$. The training set has slightly higher accuracies, with true positive probability of $\sim 77\%$ and true negative $\sim 80\%$.



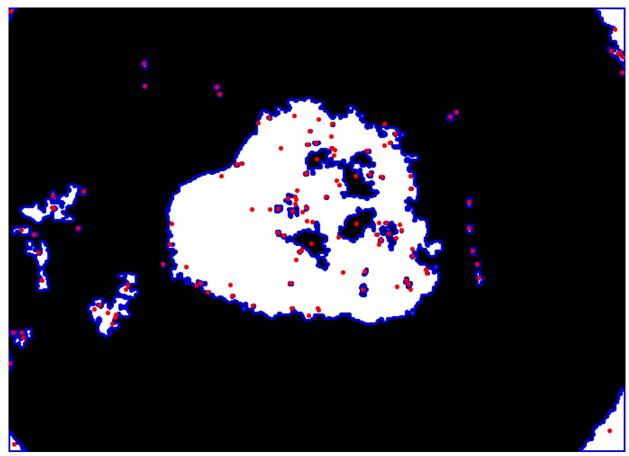
(a) Original image.



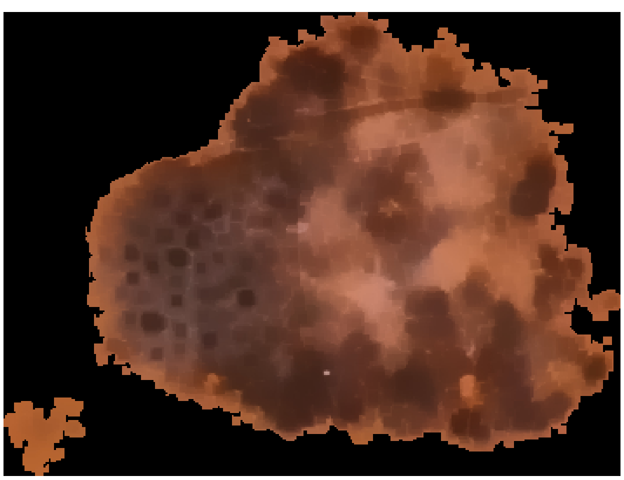
(b) Hair removal.



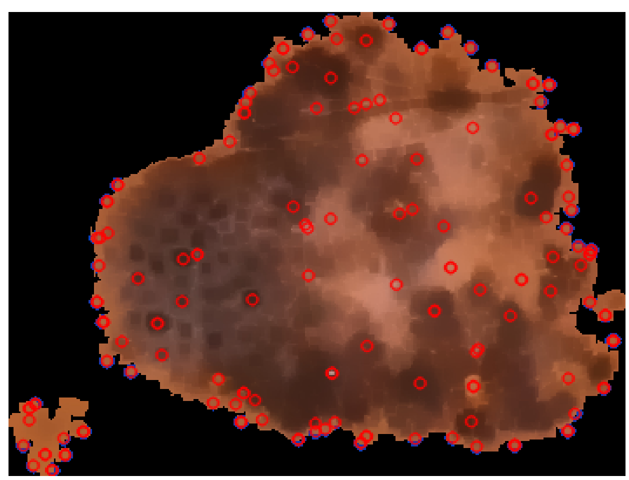
(c) CLAHE.



(d) Mask and contours.



(e) Final image.



(f) SIFT keypoints.

Fig. 1: segmentation and features extraction pipeline as described in Sec. II. In (d) the red dots represent the centroids of the detected regions. (e) is the final image with the application of the bilateral filter. The original image is ‘ISIC_0000319.jpg’ classified as a Nevus.

TABLE 1: confusion matrix of the test set. Rows represent the true labels, and the columns the predicted values. ‘MEL’ stands for ‘Melanoma’, and ‘NV’ for ‘Melanocytic Nevus’

NV	MEL
0.76	0.24
0.25	0.75

In Fig. 2 the importances of the forest are shown, where the errorbar is calculated as the standard deviation of the importances of the single trees. The most important feature appears to be the color, in particular the mean of the A channel, followed by the number of detected features.

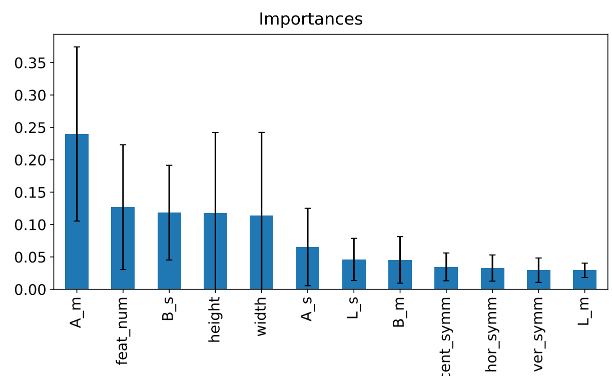


Fig. 2: importances of the best model. The capital letters stand for the LAB channels, they are followed by either an ‘m’ which stands for ‘mean’ or ‘s’ which stands for ‘standard deviation’.

Interestingly enough, letting the forest overfit by removing the limit on the maximum depth leads to a true positive probability on the validation set of $\sim 96\%$, however this model was discarded as the true negative probability is $\sim 46\%$, and the detection of a Melanoma is quite more important. The difference is possibly due to the number of samples of the Nevi in the dataset.

B. Segmentation

As mentioned in Sec. III, the algorithm has many parameters and operations which are based on arbitrary choices to address specific issues encountered in the dataset. While the majority of the images are well segmented, and the classification results are acceptable, there is still a part of the images which the algorithm fails to process. In Fig. 3 on the next page some examples are shown. The first example (a–b) fails as Otsu’s thresholding detects the transition between the bright region of the skin and the dark region of the instrument. The image is then cropped too little and almost the whole ring—with the exception of the open bottom part—is taken as a candidate region. Furthermore some fair hair is noticeable on the processed image, as the closing operation only removes darker hair.

The second one (c–d) fails for a similar reason, as the color blue is not ignored by the filter on H, and it corresponds to a large dark patch.

To process all the images the algorithm took about $70\,\mathrm{min}$ using 8 cores.

V. Conclusions

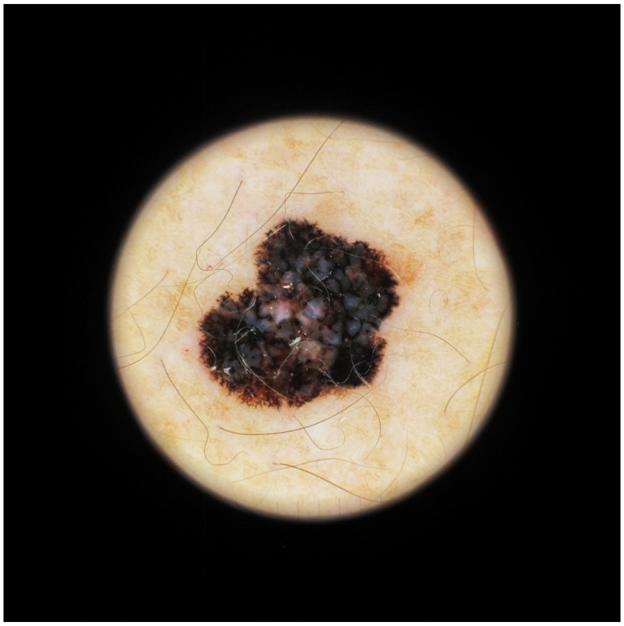
In this work ways to segment and classify skin lesion images—Melanomas and Melanocytic Nevi in particular—have been explored.

Given the nature of the images the main employed tool has been Otsu’s thresholding, leading to a fairly consistent segmentation despite not perfect application over the whole dataset (Fig. 3 on the following page). Nonetheless this is an expected behaviour since the preprocesssing algorithms require arbitrary parameters, and are not supposed to work on every kind of image.

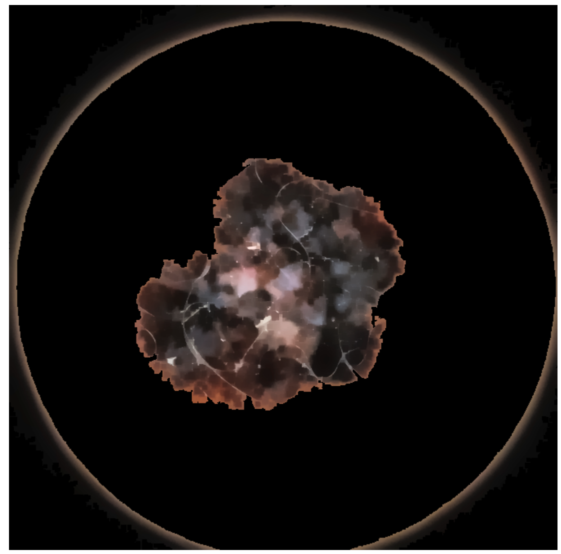
After the segmentation several features representing the ABC rule of Melanomas have been extracted and fed to a random forest algorithm, which led to an accuracy for both classes of $\sim 75\%$, proving the viability of the segmentation and the features.

The pipeline took about $70\,\mathrm{min}$ to process $17\,397$ images on 8 cores, which is arguably slow. Possibly implementing the algorithm on a GPU would result in a significant speed up.

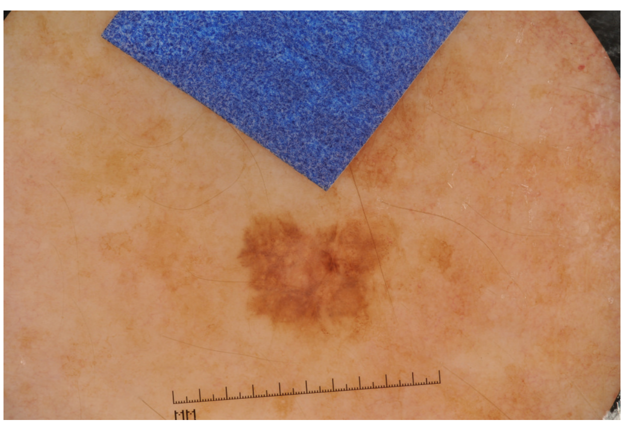
Finally more advanced techniques including deep learning architectures such as Autoencoders and Convolutional Neural Networks could be used to enhance both the segmentation and the classification.



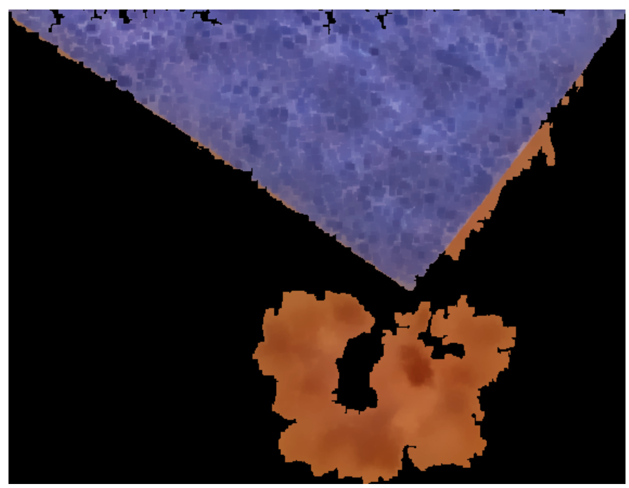
(a) Original image.



(b) Segmented image.



(c) Original image.



(d) Segmented image.

Fig. 3: the first row displays the failed segmentation of ‘ISIC_0055513.jpg’, the second row the one of ‘ISIC_0014136_downsampled.jpg’. Both Melanomas show evident artifacts.